CYTOMACHIA

κύτος → kytos → cell

μάχη → máchē → competition

Interactive 2D Cellular Automata Simulator

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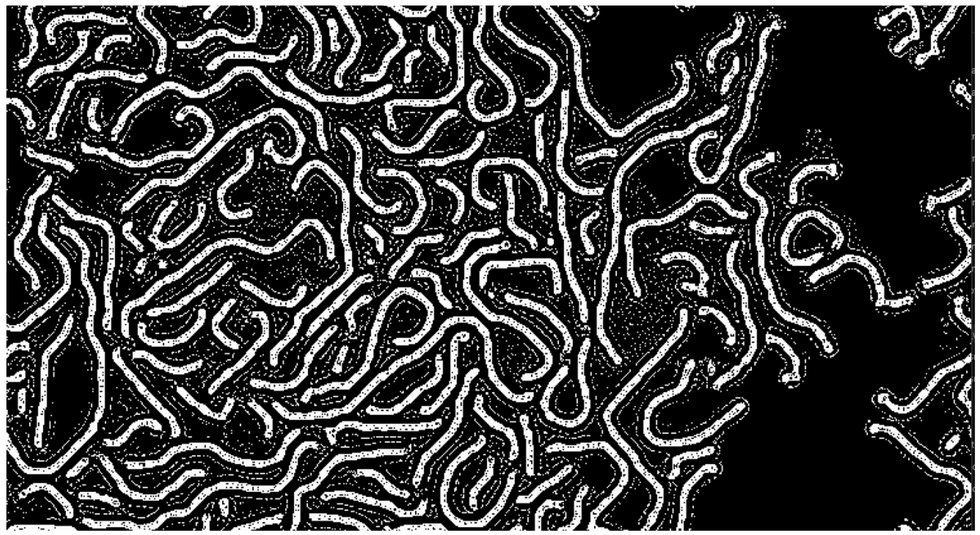
What is Cytomachia?

Cytomachia is a interactive web application to explore the world of complex 2D cellular automata simulations. Designed to both serve as an introduction to cellular automata as well as providing a platform for complex CA analysis.



Challenges

- UI for complex CA rules and kernel customization
- Writing efficient WGSL shaders for simulation
- GPU buffer setup and memory-managment to allow real-time edits
- Translating user input from typescript variables into data structures and data tyeps used by GPU buffers and shaders

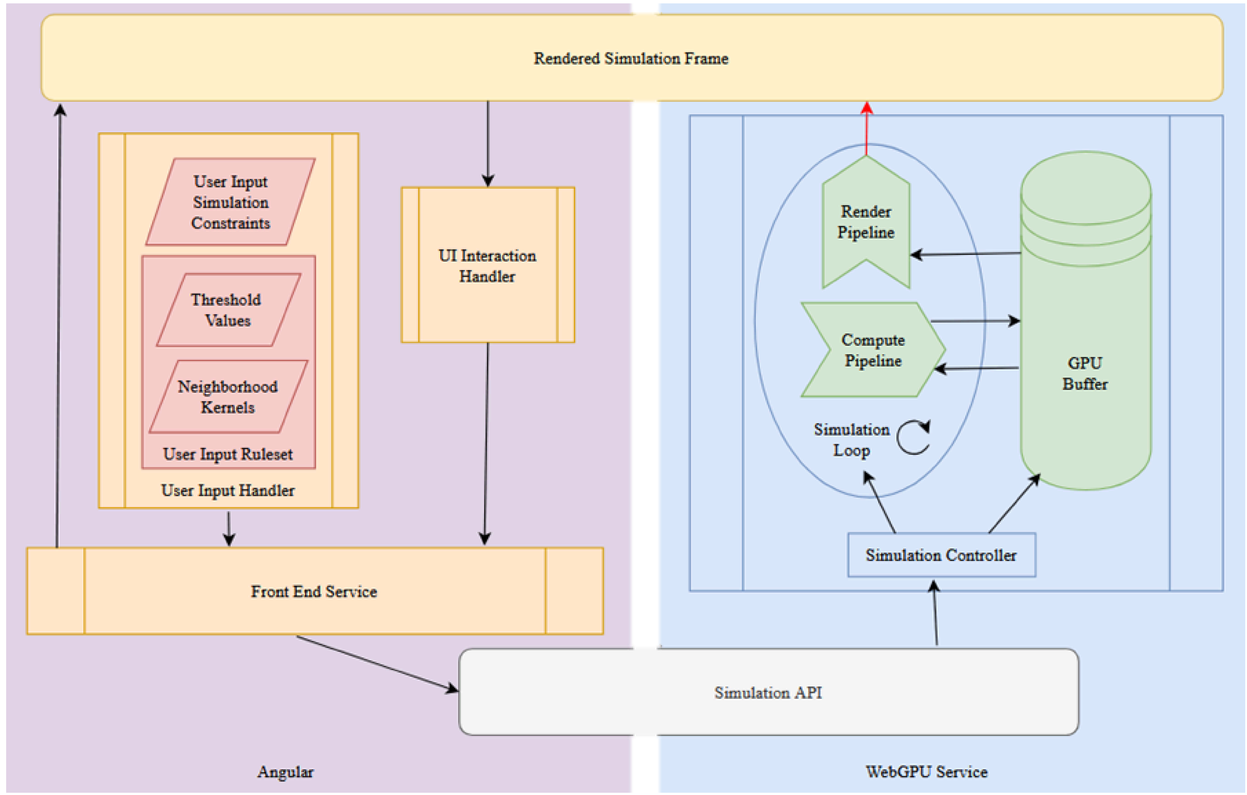
Technologies

node JS · HTML CSS JS · Angular · WebGPU

Features

- Massively configurable multi-neighborhood cellular automata simulation
- Configurable neighborhood kernels (shapes/size/weights)
- 2k simulation rendered at up to 240 FPS with GPU compute shader
- Rule customization during run-time
- Client-side compute/rendering

Cellular Automata TLDR

- 2D grid of cells
- Cells have a state, alive or dead
- Grid updated at discrete time steps
- Cells update their state based upon the conditions around them
- Simple rules result in complex and emergent behavior

System Design